Lab 2: Raytracing (2h)

CGR 2026

Instructions

This laboratory consists of five questions.

- Please ensure that you have downloaded this problem sheet during your lab slot.
- You are encouraged to read through the questions beforehand, but please leave the active solving for the session.
- Discussion is encouraged, but please write your own answers. Keep your answers concise.
- Submit handwritten answers to the tutor before the end of the lab. Show your working.
- Please **do not** nominate others to submit your work.
- Please put your name and student number on your submission.
- You may use lecture material or web resources to look up information.
- You may **not use generative AI tools** to solve or check (before submission) answers.

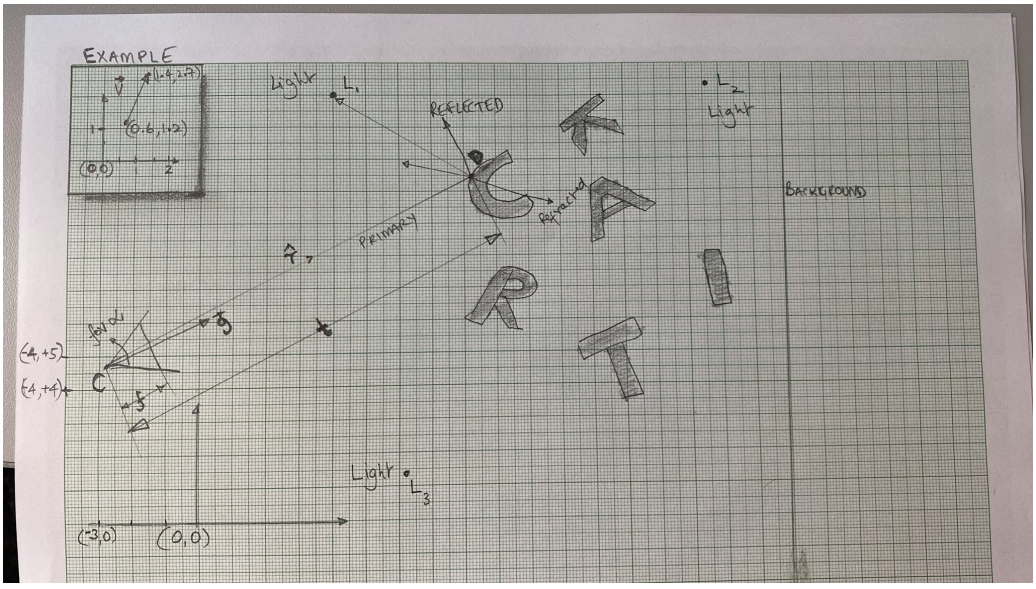


Figure 1: Example illustration of 2D scene for Question 1 on graph paper (will be supplied).

Question 1: Raytracing 2D alphabets

1. Using graph-paper, draw a scene containing the first 5 letters of your first and last names, in arbitray locations/orientations (see Figure 1). If your name has fewer than 5 letters, you may pad it with any letters of your choice. Draw each letter as an object, preferably as a block-capital with some thickness. Also draw a background plane, a pinhole camera, and at least three point lights. Use your drawing to estimate coordinates and directions. The example in Figure 1 (inset) shows a vector $\overrightarrow{v}$ for which we can read the start and end points from the graph

paper as approximately $(0.6, 1.2)$ and $(1.4, 2.7)$ and therefore the caclucate the vector and its corresponding unit vector (normalising by its length). The figure is meant as a hint towards your construction, rather than to be used as a reference for your answers. Using this approach, answer the following questions. Show all calculations and annotate your diagrams.

(a) Record the camera position C, unit gaze direction $\mathbf{g}$, field of view α, and focal distance f. Draw and label the camera's viewing direction and image plane.

(b) Draw a primary ray from the camera C through a chosen image-plane point. Show where it intersects a planar face of one of your letters at point A.

(c) Write the equation of the plane containing the selected letter face, using a point P on the plane and its normal $\mathbf{n}$ as in lecture 7. Write the primary ray as $\mathbf{R}(t) = \mathbf{C} + t\mathbf{r}$, where $\mathbf{r}$ is its unit directional vector. Solve for t, calculate the intersection point A, and check that it lies in front of the camera and within the finite face of the letter. Compare your calculated position with the approximate location on your graph paper.

(d) Choose a different point B on a letter face that lies along a primary ray. Draw and label the surface normal at B, and write down its unit normal vector $\mathbf{n}$. Calculate the mirror-reflection direction using the incident ray direction. Draw the reflected ray and indicate the direction in which it travels.

(e) Choose a third point D on a curved letter face along a primary ray. Draw the surface normal and calculate the refracted ray direction, assuming the letter material has refractive index $n_2 = 1.5$ and the surrounding medium is air with $n_1 = 1.0$. Use Snell's law, $n_1 \sin\theta_1 = n_2 \sin\theta_2$, and determine the refracted direction. Draw the incident and refracted rays, clearly indicating the angles relative to the normal.

(f) Calculate the shaded colour at point A using the Blinn–Phong illumination model. You may choose non-zero ambient, diffuse, and specular material coefficients, as well as a specular exponent. State your choices. Account for all three point lights, including their directions and distances. For each light, determine whether point A is in shadow; include only unoccluded light contributions. Show the ambient, diffuse, and specular terms separately, then combine them to obtain the final shaded colour. State any simplifying assumptions you use, such as omitting distance attenuation or clamping the final colour to a specified range.

Question 2: Convolution of a Discrete Signal

(a) Sketch a possible discrete one-dimensional function $f(x)$, defined for integer values $x = 1, 2, \ldots, 100$ that contains a combination of step changes, ramps, and smoothly varying regions. Clearly mark the locations of the step edges on your sketch. You do not need to specify exact numerical values.

(b) Consider convolving $f(x)$ with each of the following kernels:

$$K_1 = \left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right], \qquad K_2 = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right].$$

Without performing any calculations, describe the qualitative effect of each kernel on the signal. In particular, discuss how each kernel affects the step edges, ramps, and smoothly varying regions. Explain the differences between the two results.

(c) Suppose the convolution with each kernel is repeated several times, using the output of one convolution as the input to the next. Describe the expected qualitative effect on the signal in each case. Explain how the two kernels differ in their behaviour under repeated convolution.

Question 3: Texture coordinates inside a triangle

Consider a triangle ABC with 2D vertex positions $A = (0, 0)$, $B = (4, 0)$, and $C = (0, 4)$.

Each vertex has an associated texture coordinate:

$$A \to (u_A, v_A) = (-1, 0),$$
$$B \to (u_B, v_B) = (1, -1),$$
$$C \to (u_C, v_C) = (0, 3).$$

Choose any internal point P that lies inside the triangle. You may write any coordinates for P as long as it lies inside the triangle. We want to determine the texture coordinates (u_P, v_P) corresponding to this point.

(a) A point inside a triangle can be expressed as a weighted combination of its vertices: $P = \lambda_A A + \lambda_B B + \lambda_C C$, where λ_A, λ_B, and λ_C are the barycentric coordinates of P.
What condition must these three weights satisfy?

(b) Using the vertex positions and the coordinates of P:

(i) Write down the equations for the x- and y-coordinates of P in terms of its unknown barycentric coordinates.

(ii) Use these equations and the condition from part (a) to determine the three barycentric coordinates λ_A, λ_B, and λ_C.

(c) The texture coordinates at P are obtained by applying the same barycentric weights to the texture coordinates of the vertices: $u_P = \lambda_A u_A + \lambda_B u_B + \lambda_C u_C$ and $v_P = \lambda_A v_A + \lambda_B v_B + \lambda_C v_C$. Substitute the barycentric coordinates from part (b) and calculate the texture coordinates (u_P, v_P).

(d) Explain briefly why barycentric coordinates allow texture coordinates to be interpolated at any point inside a triangle.